

Grid-Connected 10.5 kWp Solar PV System: Design, Sizing, and Performance Analysis Using PVsyst

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Project Report — Grid-Connected Photovoltaic System Design

Bilahri C7, India

Abstract—This report presents the design, electrical sizing, and simulated performance evaluation of a 10.5 kWp grid-connected (on-grid) solar photovoltaic (PV) system located at Bilahri C7, India (23.1261°N, 79.9662°E). The design process begins with an estimation of the daily energy demand, followed by PV array sizing, inverter selection, string voltage verification, DC/AC cable sizing, voltage-drop analysis, and protection-device selection. The complete system, comprising 28 LG Electronics LG375N1C-A6 monocrystalline modules (7 in series $\times$ 4 parallel strings) and two SMA Sunny Tripower 5.0-3SE-40 inverters (10 kWac total, DC/AC ratio 1.05), was modelled in PVsyst V8.1.3. The simulation predicts an annual energy injection of 16,662 kWh with a specific yield of 1587 kWh/kWp/year and a Performance Ratio (PR) of 80.23%, closely corroborating the manual engineering estimates. The results confirm that the proposed configuration satisfies all electrical, thermal, and safety design constraints and is suitable for residential/commercial grid-tied deployment.

Index Terms—Grid-connected PV system, PVsyst simulation, inverter sizing, DC/AC ratio, performance ratio, specific yield, string sizing.

I. INTRODUCTION

With the increasing global demand for clean and sustainable electricity, grid-connected photovoltaic (PV) systems have become an essential solution for reducing dependence on fossil fuels and lowering greenhouse-gas emissions. The performance of a solar PV system, however, is highly dependent on proper system design, including accurate PV array sizing, inverter selection, string configuration, and voltage matching. An improperly sized system

can lead to inverter clipping, reduced energy production, higher electrical losses, and poor economic performance.

This project addresses these challenges by designing a 10.5 kWp on-grid solar photovoltaic system using the PVsyst simulation environment. The system incorporates 28 LG Electronics 375 W mono-crystalline PV modules and two SMA Sunny Tripower 5.0-3SE-40 inverters to achieve an optimised DC/AC ratio of 1.05, ensuring efficient power conversion while remaining within inverter operating-voltage limits.

The objective of this work is to design, simulate, and evaluate a 10.5 kWp grid-connected solar PV system, optimising the PV array configuration and inverter sizing to achieve maximum energy generation while complying with electrical and operational design constraints. The study further analyses the annual energy yield, Performance Ratio (PR), system losses, and overall efficiency of the system under realistic meteorological conditions.

II. SYSTEM DESCRIPTION

The proposed system is a fixed-tilt, grid-connected photovoltaic power plant. Solar irradiance incident on the PV array is converted into DC electricity, which is processed by grid-tied string inverters employing Maximum Power Point Tracking (MPPT) and exported to the utility grid through an AC distribution board (ACDB) equipped with protection devices. Table I summarises the key system specifications.

TABLE I. SYSTEM SPECIFICATIONS

Parameter	Specification
System type	Grid-connected (on-grid)
Simulation software	PVsyst V8.1.3
Site location	Bilahri C7, India (23.1261°N, 79.9662°E, 396 m ASL)
PV module	LG Electronics LG375N1C-A6, 375 Wp (monocrystalline)
Total modules	28
PV array capacity	10.5 kWp
Array configuration	4 parallel strings $\times$ 7 modules in series
Module area	50.8 m ² (total); 1.813 m ² (per module)
Inverter	SMA Sunny Tripower 5.0-3SE-40
Number of inverters	2
Total inverter (AC) capacity	10.0 kW
DC/AC ratio	1.05
MPPT voltage range	150–800 V

Parameter	Specification
Maximum DC input voltage	1000 V
Grid frequency / voltage	50 Hz, 230 V (single phase, per outlet)
Fixed-plane tilt / azimuth	29.3° / 0° (south-facing)
Albedo	0.20

III. DESIGN METHODOLOGY

The system was designed following a systematic engineering approach, summarised in the six steps below.

A. Load Assessment

An initial electrical load assessment was carried out for a representative three-storey residential building to establish a baseline daily energy demand (Table II). The connected load totals 9500 W with a cumulative daily energy consumption of 52.5 kWh/day, computed as the sum of the product of rated power, quantity, operating hours, and diversity factor for each load category.

B. Site and Meteorological Data

Geographical and meteorological parameters for Bilahri C7, India (23.1261°N, 79.9662°E, altitude 396 m, UTC+5.5) were sourced from the PVsyst Meteororm 9.0 database, including annual global horizontal irradiation, diffuse irradiation, and ambient temperature.

C. PV Module Selection

LG Electronics LG375N1C-A6 monocrystalline modules (375 Wp) were selected for their high conversion efficiency, proven reliability, and voltage compatibility with the chosen inverter.

D. PV Array Design

The array was configured as 4 parallel strings of 7 series-connected modules (28 modules total, 10.5 kWp), balancing string voltage against inverter MPPT limits while minimising mismatch losses.

E. Inverter Selection

Two SMA Sunny Tripower 5.0-3SE-40 inverters (5 kW each, 10 kW total) were selected, yielding a DC/AC ratio of 1.05 — within the recommended design range of 1.0–1.2.

F. Simulation

The complete system was modelled in PVsyst to evaluate annual energy production, system losses, inverter performance, Performance Ratio, and array/voltage compatibility under realistic operating conditions.

TABLE II. LOAD CALCULATION — BASELINE DAILY ENERGY DEMAND

Floor	Load	Qty	Rating (W)	Hrs/day	D.F.	Conn. Load (W)	Energy (Wh/day)
Ground	LED Lights	4	50	7	1.00	200	1400
First	LED Lights	4	50	7	1.00	200	1400
Second	LED Lights	4	50	7	1.00	200	1400
Utility	Water Pump	2	1000	1.5	1.00	2000	3000
Kitchen	Refrigerator	2	200	24	1.00	400	9600
Laundry	Washing Machine	1	1350	2	1.00	1350	2700
Living	Domestic Appliances	1	1200	1	1.00	1200	1200
Bedrooms	Air Conditioner	2	1500	6	1.00	3000	18000
All Floors	Ceiling Fans	10	75	12	1.00	750	9000
Standby	Standby Loads	1	200	24	1.00	200	4800
—	TOTAL	—	—	—	—	9500	52500 (52.5 kWh/day)

IV. DESIGN CALCULATIONS

A. Required PV Capacity

Using the baseline daily energy demand of 52.5 kWh/day and a regional average of 4.5 peak sun hours (PSH):

$$P_1 = 52.5 / 4.5 = 11.66 \text{ kW}$$

Accounting for an overall system efficiency of 77% (wiring, temperature, inverter, and mismatch losses):

$$P_2 = 11.66 / 0.77 = 15.15 \text{ kW}$$

Based on the project budget, the design target was set to offset 70% of the estimated demand:

$$P_{req} = 15.15 \times 0.70 = 10.60 \text{ kW}$$

This closely matches the finally adopted array capacity of 10.5 kWp, confirming the design target is met.

B. PV Array Sizing

With a 375 W module rating and a target capacity of 10.5 kWp:

$$N = 10,500 / 375 = 28 \text{ modules}$$

A preliminary sizing exercise explored a 4-series $\times$ 7-parallel arrangement, giving a string voltage of only 122.8 V — unnecessarily low relative to the inverter's 150–800 V MPPT window. The configuration was therefore revised and finalised as 7 modules in series $\times$ 4 parallel strings, which was carried forward into the PVsyst model and confirmed by the simulation's single-line diagram.

Per-module electrical parameters at STC: $V_{mpp} = 30.7$ V, $V_{oc} = 45.6$ V, $I_{mpp} = 42.5$ A. Module physical area = $1.74 \times 1.042 = 1.813$ m²; total array area = $28 \times 1.813 = 50.76$ m² (PVsyst reports 50.8 m², consistent within rounding).

C. Inverter Sizing

Two SMA Sunny Tripower 5.0-3SE-40 inverters were selected:

$$P_{AC} = 2 \times 5.0 = 10.0 \text{ kW}$$

$$DC/AC = 10.5 / 10.0 = 1.05$$

A DC/AC ratio of 1.05 lies within the recommended range of 1.0–1.2, ensuring efficient inverter utilisation with minimal clipping. Each inverter accepts 4 MPPT inputs; in the PVsyst model these are configured with no power sharing between MPPTs (independent-MPPT mode), which is the conservative option appropriate when each MPPT serves an equal single string, avoiding unexpected overload losses.

D. String Voltage Verification

With 7 modules in series per string:

$$V_{string(mpp)} = 7 \times 30.7 = 214.9 \text{ V}$$

$$V_{string(oc)} = 7 \times 45.6 = 319.2 \text{ V}$$

PVsyst's temperature-corrected verification confirms: $V_{mpp}(60^\circ\text{C}) = 215$ V, $V_{mpp}(20^\circ\text{C}) = 247$ V, and $V_{oc}(-10^\circ\text{C}) = 319$ V — all

safely within the inverter's 150–800 V MPPT range and well below the 1000 V maximum DC input limit. Since $150 \text{ V} < 214.9 \text{ V} < 800 \text{ V}$ and $319.2 \text{ V} < 1000 \text{ V}$, the design is fully voltage-compatible.

E. DC Cable Sizing

Maximum per-string design current, with a 1.25 safety factor:

$$I_{design} = 45.4 \times 1.25 = 56.75 \text{ A}$$

A 10 mm² copper solar DC cable (current rating $\approx$ 65 A, 1000 V DC rated) was selected, comfortably exceeding the 56.75 A design current.

F. AC Cable Sizing

At 230 V, 0.99 power factor, and 10 kW total inverter output:

$$I_{AC} = 10,000 / (230 \times 0.99) = 43.9 \text{ A}$$

$$I_{design} = 43.9 \times 1.25 = 54.9 \text{ A}$$

A 10 mm² copper XLPE cable (rating $\approx$ 60 A) satisfies this requirement.

G. Voltage-Drop Verification

For a 20 m cable run carrying 45 A with resistance 1.83 Ω /km:

$$V_d = (2 \times 45 \times 20 \times 1.83) / 1000 = 3.29 \text{ V} \text{ (1.43\%)}$$

A voltage drop of 1.43% is well below the 3% permissible limit.

H. Protection Devices

- DC fuse: 63 A DC
- DC isolator: 1000 V, 63 A
- AC breaker: 63 A, 3-pole MCB
- Surge protection: Type II SPD (DC, 1000 V) and Type II SPD (AC, 230 V)
- Earthing: dedicated copper earthing system

These devices jointly protect against overcurrent, short-circuit, reverse-current, lightning surge, and grid-fault conditions.

A. Project and System Summary

The simulation reports an installed PV capacity of 10.50 kWp, total inverter AC capacity of 10 kWac (Pnom ratio 1.05), and unlimited grid load conditions. The array operates at 50°C with $P_{mpp} = 9.63$ kW, $U_{mpp} = 223$ V, and $I_{mpp} = 43$ A.

V. PVSYST SIMULATION RESULTS

The finalised configuration (28 $\times$ LG375N1C-A6 modules, 4 strings $\times$ 7 series, two SMA Sunny Tripower 5.0-3SE-40 inverters) was simulated in PVsyst V8.1.3 using an hourly time step, Perez transposition, and Meteororm 9.0 synthetic weather data for Bilahri C7, India, with no shading scenario defined.

TABLE III. PVSYST RESULTS SUMMARY

Parameter	Result
Produced / injected energy	16,662 kWh/year
Specific production	1587 kWh/kWp/year
Performance Ratio (PR)	80.23 %
Simulation step	Hourly
Number of PV modules	28
Number of inverters	2

TABLE IV. MONTHLY IRRADIATION, TEMPERATURE, AND ENERGY BALANCE (PVSYST)

Month	GlobHor (kWh/m ²)	T_Amb (°C)	GlobInc (kWh/m ²)	EArray (kWh)	E_Grid (kWh)	PR
Jan	133.3	17.00	182.6	1702	1598	0.833
Feb	142.3	20.80	178.5	1628	1531	0.817
Mar	188.2	26.00	208.7	1861	1752	0.800
Apr	201.4	31.20	198.3	1731	1628	0.782
May	218.8	34.70	197.1	1701	1597	0.772
Jun	163.7	31.70	143.2	1275	1184	0.787
Jul	111.4	27.50	98.9	899	817	0.786
Aug	104.0	26.60	96.8	886	803	0.790
Sep	137.6	27.30	142.0	1285	1194	0.801
Oct	155.6	25.90	182.2	1636	1533	0.802
Nov	130.7	21.80	170.1	1562	1463	0.819
Dec	127.3	18.20	179.6	1665	1562	0.828
Year	1814.0	25.74	1977.9	17,832	16,662	0.802

B. Loss Diagram

The PVsyst loss diagram traces the conversion from horizontal irradiation to grid-injected energy: 1814 kWh/m² global horizontal irradiation is increased by 9.0% on the 29.3° tilted plane and reduced by 1.99% for IAM (Fresnel/AR-coating) effects, yielding 1939 kWh/m² of effective irradiation over the 51 m² collector area. At a

module STC efficiency of 20.74%, this corresponds to an array nominal energy of 20,414 kWh. Successive losses of −0.58% (low-irradiance), −10.00% (thermal), +0.75% (module quality), −2.00% (mismatch), and −1.12% (ohmic wiring) yield an array virtual energy at MPP of 17,832 kWh. Inverter losses of −4.34% (efficiency), −0.09% (power threshold), and −2.23% (night consumption) result in a final available/injected energy of 16,662 kWh/year.

VI. PERFORMANCE ANALYSIS

A. Performance Ratio

The engineering (manual) estimate uses:

$$PR = \text{Final Yield} / \text{Reference Yield} = 4.4 / 5.3 = 83\%$$

PVsyst's hourly simulation reports a PR of 80.23%, closely matching the manual estimate; a Performance Ratio above 80% indicates an efficiently designed system with relatively low system losses.

B. Capacity Factor

$$CF = \text{Annual Energy} / (\text{Installed Capacity} \times 8760)$$

Manual estimate (17,850 kWh, 10.5 kW): CF = 19.4%. Using the PVsyst-simulated annual energy of 16,662 kWh: CF = 16,662 / (10.5 × 8760) = 18.1%. Both values are typical for a well-designed grid-connected PV system under central-Indian climatic conditions.

C. Specific Yield

$$SY = \text{Annual Energy} / \text{Installed Capacity}$$

Manual estimate: SY = 17,850 / 10.5 = 1700 kWh/kWp/year. PVsyst-simulated: SY = 16,662 / 10.5 = 1587 kWh/kWp/year. Table V compares the manual engineering estimates against the PVsyst simulation outputs.

TABLE V. MANUAL ESTIMATE VS. PVSYST SIMULATION

Metric	Manual Estimate	PVsyst Simulation
Annual energy	17,850 kWh	16,662 kWh
Specific yield	1700 kWh/kWp/yr	1587 kWh/kWp/yr
Performance Ratio	83.0 %	80.23 %
Capacity factor	19.4 %	18.1 %

VII. DISCUSSION

The close agreement between manual engineering calculations and PVsyst simulation results (Table V) validates the design methodology adopted in this project. The small deviation (approximately 6–7%) between the manual specific-yield estimate and the simulated value is attributable to PVsyst's more detailed hourly modelling of thermal losses (−10%), night-time inverter

consumption (−2.23%), and IAM effects, which are only approximated in the manual calculation. The DC/AC ratio of 1.05 keeps inverter clipping losses negligible (0.00% loss over nominal inverter power in the PVsyst report), while all string voltages remain safely within the 150–800 V MPPT window across the full ambient temperature range considered (−10°C to 60°C). The selected 10 mm² DC and AC cables, together with the specified protection devices,

satisfy current-carrying capacity and voltage-drop ($1.43\% < 3\%$) requirements with adequate margin.

VIII. FUTURE SCOPE

- Integration of Battery Energy Storage Systems (BESS) for backup and peak-shaving.
- AI/machine-learning-based solar power forecasting.
- Real-time monitoring via IoT and SCADA platforms.
- Economic analysis including Levelized Cost of Energy (LCOE).
- Carbon-emission reduction assessment.
- Digital-twin implementation for predictive maintenance.
- PV degradation modelling over a 25-year project lifetime.
- Incorporation of bifacial PV modules.
- Layout optimisation using genetic algorithms or particle-swarm optimisation.

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IX. CONCLUSION

This project has presented the complete design, sizing, and PVsyst-based performance evaluation of a 10.5 kWp grid-connected solar PV system for Bilahri C7, India. Starting from a baseline daily energy demand of 52.5 kWh/day, the required PV capacity was derived (10.60 kW) and closely matched by the final adopted array of 28 LG375N1C-A6 modules (7 series $\times$ 4 parallel, 10.5 kWp). Two SMA Sunny Tripower 5.0-3SE-40 inverters (10 kW total, DC/AC ratio 1.05) were verified to be voltage-compatible and appropriately sized. Cable sizing, voltage-drop, and protection-device selection were all confirmed to satisfy applicable design margins. PVsyst simulation predicts an annual grid-injected energy of 16,662 kWh, a specific yield of 1587 kWh/kWp/year, and a Performance Ratio of 80.23%, corroborating the manual engineering estimates and confirming that the proposed system is technically feasible, efficient, and suitable for grid-tied residential/commercial deployment.